EKE



Technical Specifications

Smarter trains. Better future.

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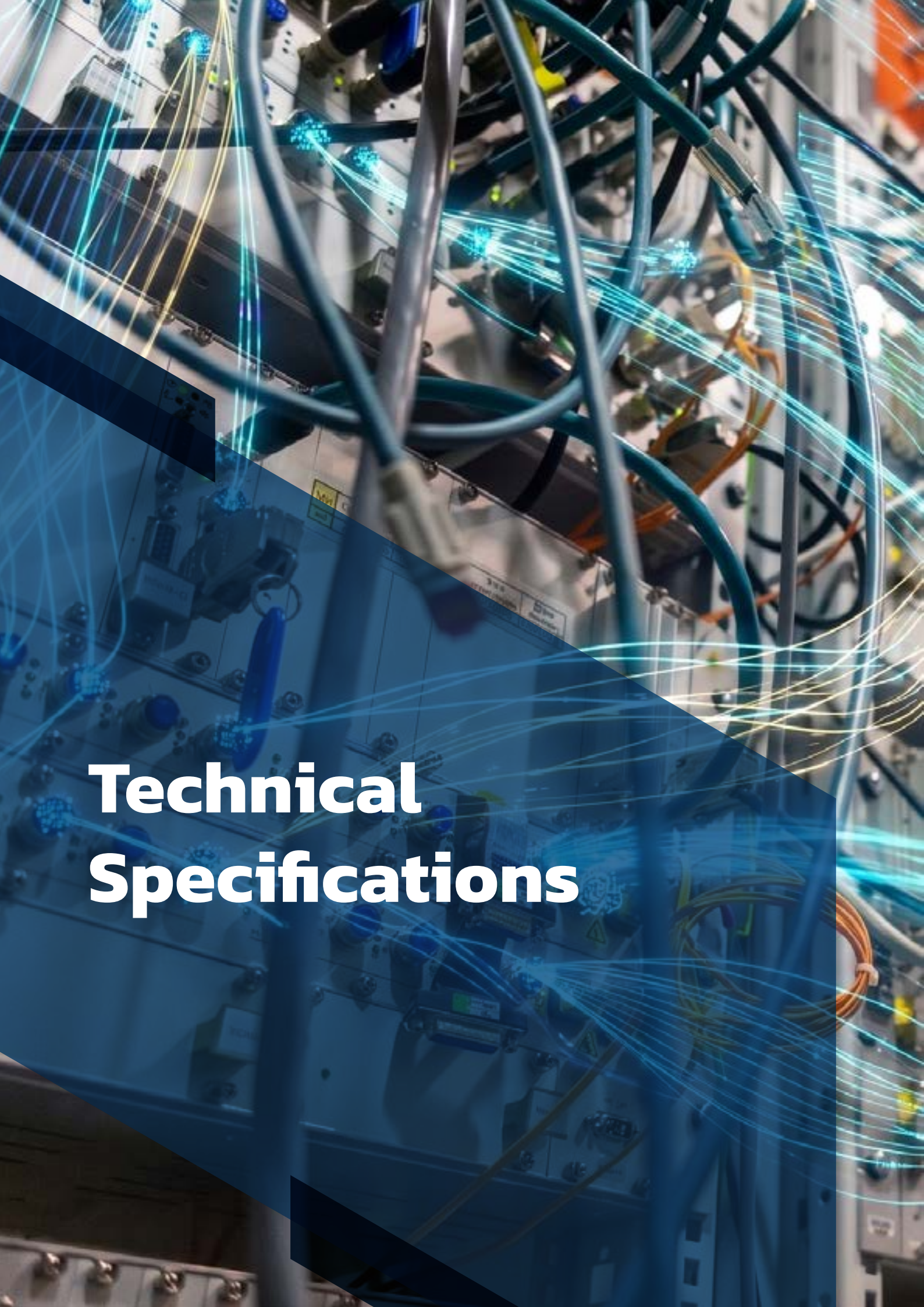
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Technical Specifications

Central Processing Unit 6000 Series

The CPU6000 Series offers a scalable selection of processing options, from high performance multicore units to cost optimized single core modules. With multiple performance variants, you can select exactly the processing capability each system requires. This avoids unnecessary cost or over specification while ensuring the system remains efficient, scalable and well matched to the requirements of different train designs.

Key Features

Secure Boot and Cybersecurity Compliance

A cryptographically verified secure boot chain ensures that only trusted, manufacturer-approved software runs on a device. Meets railway-specific requirements defined in IEC 62443-4-2 to Security Level 2. The architecture provides the capability to run intrusion detection agent software, enabling proactive threat monitoring and enhanced system protection.

SDTv2 and TRDP Communication Support

For safety-critical applications, the system supports SDTv2 (Safe Data Transmission), ensuring safe and dependable data exchange. It also implements the Train Real-Time Data Protocol (TRDP), the standard network protocol for IP-based communication, enabling interoperability and high performance connectivity.

CRA Ready

Secure-by-design architecture developed in accordance with the Cyber Resilience Act (CRA), Regulation (EU) 2024/284 supporting long-term cybersecurity and lifecycle maintenance requirements.

SIL 2 Certified

Designed and certified for safety-related functions up to SIL 2 according to EN 50129 and EN 50716, helping reduce system certification effort and supporting compliance with railway safety requirements.

Network Redundancy with Dual Homing

Dual homing as standard, allowing the device to maintain connections to two independent network paths. This architecture increases system availability and communication resilience by supporting continued operation in the event of a network fault or link failure.

GoA4 Automated Train Operation

Capable of Grade of Automation (GoA) up to Level 4 in compliance with EN 62290. The CPU6000 Series provides the processing performance, communication capabilities and cybersecurity features required for highly automated train operations, supporting system architectures ranging from driver-assisted operation through to unattended train operation.



EKE-Trainnet® CPU6000



EKE-Trainnet® CPU6010



EKE-Trainnet® CPU6030

Compliance

- EN 50155
- EN 45455-2
- EN 50716
- EN 50126
- EN 50129
- EN 62290
- IEC 61131
- IEC 61375
- IEC 62443-4-2
- IEC 62425
- CLC/TS 50701
- SIL 2

Technical Specifications

Dimensions (W x H x D):
8 TE x 3 U x 160 mm (CPU6000)
4 TE x 3 U x 160 mm (CPU6010/CPU6030)

Weight:
295 g (CPU6000)
193 g (CPU6010/CPU6030)

Temperature Range (operational):
-40 °C...+70 °C

Power Consumption:
5 W (CPU6000)
2 W (CPU6010/CPU6030)

MTBF (40 °C ambient temperature):
> 1 000 000 h

Central Processing Unit:
2 GHz multi-core ARM processor with 64-bit instruction set (CPU6000)
1 GHz multi-core ARM processor with 64-bit instruction set (CPU6010)
400 MHz multi-core ARM processor with 64-bit instruction set (CPU6030)

Ethernet Interfaces:
3 x 1 Gbps, connector type M12, X-coded (CPU6000)
2 x 100 Mbps, connector type M12, D-coded (CPU6010/CPU6030)

Serial Console:
USB-C

File System & Event Logging Flash Memory:
16 GB eMMC Flash

Processor RAM:
2 GB LPDDR4 SDRAM, speed 2666 MT/s (CPU6000)
1 GB LPDDR4 SDRAM, speed 1600 MT/s (CPU6010/CPU6030)

Front panel memory card slot:
SD Card for firmware updates, and device recovery (CPU6000)

Real-Time Clock With Backup Power:
30 days. Maintains an accuracy better than 10 seconds over 50 days under room temperature conditions.

Wire Train Bus Interface Module

EKE-Trainnet® Wire Train Bus (WTB) modules are used to create train-wide communication networks. They implement the WTB link layer functions of the IEC 61375-2-1 Train Communications Network (TCN) standard. The TCN's real-time protocols, the UIC leaflet 556 specifications and the routing between the WTB and other buses are implemented by the gateway Central Processing Unit (CPU) module.

Compliance

- EN 50155
- EN 45545
- IEC 61375-2-1

Key Features

Supports Physical Cable Redundancy

Physical cable redundancy as specified in the IEC 61375-2-1 standard is supported.

Efficient Processing with Diagnostics

The WTB module has the capability of carrying out enhanced diagnostics functions, like monitoring the bit error ratios between individual train coaches in order to identify communication issues, in compliance with IEC 61375-1.

Static Shared RAM Memory

Static shared RAM memory accessible from both the local CPU and the VME Bus interface. The TCN standard's source and sink port handling for WTB process data is supported. A separate memory area is reserved for message data and maintenance messages as well as for Remote Procedure Call type access to the Link Layer primitives. The WTB module is able to provide applications with timing synchronisation when a Macro Period finishes.

Intelligent Repeater Management

In order to handle incompatible cabling or extremely harsh operating conditions, the WTB module is able to act as a bus repeater by completely regenerating the data streams towards both extremities. It supports the selection of an optimal number of concurrently enabled repeaters on the train bus, thus allowing the network timing to retain undegraded specifications.

Options

Fritting Voltage

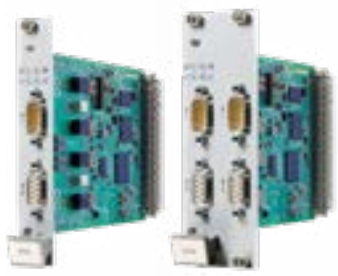
The EKE-Trainnet® WTB bus interface sustains fritting voltage via WTF2259, WTF2376 and WTF1765 modules.

Separated Connectors for the Redundant Bus

Separated connectors for the redundant bus are available as an option via WTB1912, WTF2376 and WTF1765 modules.

Coded Bus Connectors

Coded D-connectors are available as an option via the WTF1765 module.



EKE-Trainnet® Wire Train Bus Interface Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm (WTB1822)
8 TE x 3 U x 160 mm (WTB1912)
8 TE x 3 U x 160 mm (WTF1765)
4 TE x 3 U x 160 mm (WTF2259)
8 TE x 3 U x 160 mm (WTF2376)

Weight:
166 g (WTB1822)
214 g (WTB1912)
226 g (WTF1765)
177 g (WTF2259)
225 g (WTF2376)

Input Power:
5 V DC ± 5 % (1 A max., 0.5 A typ.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
1 370 000 h (WTB1822)
1 330 000 h (WTB1912)
1 010 000 h (WTF1765)
1 030 000 h (WTF2259)
1 010 000 h (WTF2376)

Train Bus:
IEC 61375-1: WTB

Data Rate:
1 Mbit/s (Half speed 500 kbit/s on request)

MAU:
Transformer isolation
Combined or individual bus connectors

VME Bus (IEC 821) Interface:
A24 Slave with D08(EO)/D16
256 kB DP RAM for process and message data

Multifunction Vehicle Bus Interface Module

EKE-Trainnet® Multifunction Vehicle Bus (MVB) modules are used to create vehicle buses or more comprehensive train communication networks. The MVB module implements the MVB link layer functions of the IEC 61375-3-1 Train Communications Network (TCN) standard. The TCN's real-time protocols and the routing between the MVB and other buses are implemented by the gateway Central Processing Unit (CPU) module.

Compliance

- ✓ EN 50155
- ✓ EN 45545
- ✓ IEC 61375-3-1

Key Features

Supports Physical Cable Redundancy

Physical cable redundancy and full bus master functionality as specified in the IEC 61375-3-1 standard is supported. Bus administrator redundancy is supported on the software level.

Static Shared RAM Memory

Static shared RAM memory accessible from both the local CPU and the VME Bus interface. A separate memory area is reserved for message data and maintenance messages as well as for Remote Procedure Call type access to the Link Layer primitives.

MVB Communication and Control

The MVB link layer functions are implemented by the MVB module's local processor. Multiple EKE-Trainnet® MVB modules can be controlled by a single CPU to implement physically separated MVB networks.

Efficient Processing and Diagnostics

The MVB module has the capability of carrying out some diagnostic functions in compliance with IEC 61375-3-1.

Options

Physical Interface

The MVB module offers the following physical interfaces:

- ✓ Electrical Short Distance with isolation (ESD+)
- ✓ Electrical Medium Distance (EMD).

Each are equipped with two sub D-9 connectors.



EKE-Trainnet® Multifunction Vehicle Bus Interface Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
160 g (EMD, MVB1820)
152 g (ESD+, MVB2791)

Input Power:
5 V DC ± 5 % (1 A max., 0.5 A typ.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
1 650 000 h (EMD, MVB1820)
1 580 000 h (ESD, MVB2791)

Physical Medium:
Electrical Medium Distance (EMD, MVB1820)
Isolated Electrical Short Distance (ESD+, MVB2791)

Data Rate:
1.5 Mbit/s; Manchester encoding

VME Bus (IEC 821) Interface:
A24 Slave with D08(EO)/D16
246 kB SRAM and 512 kB traffic memory for process and message data

CAN Vehicle Bus Interface Module

The EKE-Trainnet® CAN Vehicle Bus Interface Module (CVB) is used to create up to two CAN buses in the train. The two CAN ports can be independently configured as CAN 2.0 A/B or CANopen®. The CVB module implements the CAN link layer functions. The routing between the CAN bus and other buses is implemented by the gateway Central Processing Units (CPU) module.

Compliance

- EN 50155
- EN 45545
- IEC 61375-3-3
- SAE J1939

Key Features

Dual Independent CAN Network Interfaces

Two CAN 2.0 B and ISO 11898 compatible network connectors on the front panel, with one CiA DS-102 standard pin. 9-pin sub-miniature D-type connector for each independent interface. Communication to the gateway is done through shared memory over the IEC 821 VME back plane bus. Multiple CVB modules can be controlled by a single CPU to implement physically separated CAN networks.

Static Shared RAM Memory

Static shared RAM memory accessible from both the local CPU and the VME Bus interface.

Efficient Processing And Diagnostics

The CVB Interface Module has the capability of carrying out some specific diagnostics functions.

Options

Integrated Bus Termination Resistors

The CVB1621 module has terminated CAN interfaces, whereas the CVB2456 module has non-terminated CAN interfaces.



EKE-Trainnet® CAN Vehicle Bus Interface Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
149 g

Input Power:
5 V DC ± 5 % (500 mA nominal)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
1 880 000 h (CVB1621 CAN terminated)
1 880 000 h (CVB2456 CAN not terminated)

CAN Network Interface:
Two CAN interfaces

Data Rate:
Up to 1 Mbit/s

Protocol:
CAN 2.0 compatible

Electrical:
24 V compliant, optically isolated

Connector:
9-pin sub-miniature D-type connector

VME Bus (IEC 821) Interface:
A24 Slave with D08(EO)/D16

Serial Links Interface Unit

The EKE-Trainnet® Serial Links Interface Unit (SIU) is used to implement up to eight serial links. The physical layer level and parts of data link layer are implemented within SIU and higher protocol layers are implemented on the system's Central Processing Unit (CPU). For gateway applications, the routing between the Serial Link buses and other buses are implemented by the train computer CPU module.

Compliance

- EN 50155
- EN 45545

Key Features

Configurable Serial Links

The SIU has one DIN41612- F48 connector on the front panel where Serial Links can be connected. Serial links are split into two identical groups which are isolated from the FPGA and from each other. In each group there are four Serial Links that can be configured in different ways.

Double Buffered Serial Links

The SIU provides 2 kB long RX and TX double buffers for each serial link. The system CPU can access these buffers through the VME shared memory. A dedicated FPGA controls precise timing for all eight channels in parallel and provides convenient double buffering for the system CPU. This guarantees high performance in demanding safety critical applications.

Module & Train Computer/ System Communication

Communication between the module and the train computer/system is done through shared memory over the IEC 821 VME back plane bus.

Options

Optional Handshake Signals

Optional handshake signals are available with RS-232 (SIU3449).



EKE-Trainnet® Serial Links Interface Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
140 g

Input Power:
5 V DC ± 5 % (250 mA typ., 400 mA max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
2 600 000 h

Connector:
DIN41612-F48 (at front)

Serial Link Interfaces:
2 isolated groups

Isolation Group A:
4 x RS-485 (SIU3448)
4 x RS-485 (SIU3449)

Isolation Group B:
4 x RS-485 (SIU3448)
2 x RS-232 + 2 x RS-485 (SIU3449)

FPGA Shared Memory:
TX buffers: 8 x 2 x 2 kB
RX buffers: 8 x 2 x 2 kB

VME Bus (IEC 821) Interface:
A24 Slave with D08(EO)/D16

RS-485 Half or Full Duplex
8 ports (SIU3448)
6 ports (SIU3449)
Any two RS-485 ports can be used for a single RS-442 application

RS-232
2 ports (SIU3449)

Integrated Ethernet Train Backbone Node (ETBN)

The Fast Ethernet Integrated Train Backbone Node (ETBN) (Gen2), from our partner VDS, enables the connection of networks across two or more trains. It is designed in accordance with IEC 61375 requirements for Ethernet Train Backbone (ETB) networks and is fully EN 50155 compliant. Built for the demanding conditions of on-board environments, it delivers reliable and robust performance for train systems.

Compliance

- EN 50155
- IEC 61375
- EN 50121-3-2
- IEC 60068-2-1
- IEC 60068-2-2
- EN 60068-2-30
- EN 61373
- EN 50124-1
- EN 45545-2

Key Features

Network Implementation

There are four 100BASE-TX switched ports for Train Backbone Network implementation and ten 100BASE-TX switched ports for the Consist Network implementation.

Bypass Function

Two pairs of Backbone ports are equipped with bypass function, one pair of Consist ports can be optionally equipped with bypass function.

Train Inauguration Procedure

IEC 61375 Train inauguration procedure is fully supported as well as ETBN control interface for TRDP devices. Routing and address translation rules for multiple consist networks interconnections are automatically defined during train inauguration. Other interfaces (SNMP, REST) are also available in order to provide a wider range of potential implementations.

Automatic Device Configuration

The TTCMP© services provide automatic device configuration and continuous monitoring giving the customer a way to reduce commissioning and maintenance costs.



VDS Integrated Ethernet Train Backbone Node

Technical Specifications

Dimensions (W x H x D):
207 x 184 x 51 mm
(without connectors)

Weight:
2.2 Kg ± 5% (noPoE version)
2.3 Kg ± 5% (PoE version)

Temperature Range (operational):
Standard:
-25 °C...+70 °C (+85 °C for 10 min)
Optional:
-40 °C...+70 °C (+85 °C for 10 min)

Fast Ethernet Connectors:
M12, female, 4-ways, D-coding

Power Supply Connector:
M12, male, 4-ways, A-coding

Maintenance Port Connector:
M12, female, 8-ways, A-coding

Power Supply Voltage (insulated):
Nominal 24 - 110 V
LVPS 28.5/37.5/57.0/74.0 V DC

Interruption Voltage Supply Class:
S2

PoE Class:
0, 1, 2, 3 and 4 (max 30W per port)

PoE Budget (option):
60 W derating of 1 W/ °C > 55 °C

Total Power Consumption
22 W (noPoE version)
100 W (PoE version)

Degree of protection
Standard IP40
Optional IP65

Consist Switch

The 14-port Fast Ethernet Consist Switch (Gen2), from our partner VDS, is an EN 50155-compliant Layer 2 managed switch, designed for network applications in train environments. It operates reliably in the harsh conditions typical of on-board use, providing the robustness and dependability required for train systems.

Compliance

- EN 50155
- EN 50121-3-2
- IEC 60068-2-1
- IEC 60068-2-2
- EN 60068-2-30
- EN 61373
- EN 50124-1
- EN 45545-2

Key Features

Network Implementation

There are fourteen 100BASE-TX ports. Two pairs of Fast Ethernet ports can be optionally equipped with bypass function. Eight Fast Ethernet ports can be optionally equipped with PoE+ capability.

Layer 3 Capabilities

Layer 3 capabilities allow IPv4 unicast and multicast routing among different VLANs. Its purpose is to enable the implementation of network topologies in a train.

Advanced Traffic Switching Capabilities

Advanced traffic switching capabilities include support for VLANs, IGMP, Spanning Tree protocols (STP/RSTP) and the possibility to apply Quality of Service (QoS) policies. Layer 3 extensions include inter-VLAN routing with packet filtering together with NAT. OSPF and RIP dynamic protocols are also supported. IPv4 router redundancy can be provided through standard VRRP protocol. performance.

On-Board Network Discovery

The device includes a sophisticated programming interface and is capable of completely auto-configure itself from an abstract project definition right after power on. Both these features dramatically ease maintenance activities and provide smart tools for fine-grained control over device operations and configuration.

Bypass Feature

The bypass feature makes the device completely transparent to the network in case of power loss or failure, thus avoiding or reducing the impact of possible local dropouts or malfunction on network



VDS Consist Switch

Technical Specifications

Dimensions (W x H x D):
207 x 184 x 51 mm
(without connectors)

Weight:
2.2 Kg ± 5% (noPoE version)
2.3 Kg ± 5% (PoE version)

Temperature Range (operational):
Standard:
-25 °C...+70 °C (+85 °C for 10 min)
Optional:
-40 °C...+70 °C (+85 °C for 10 min)

Fast Ethernet Connectors:
M12, female, 4-ways, D-coding

Power Supply Connector:
M12, male, 4-ways, A-coding

Maintenance Port Connector:
M12, female, 8-ways, A-coding

Power Supply Voltage (insulated):
Nominal 24 - 110 V DC
LVPS 28.5/37.5/57.0/74.0 V DC

Interruption Voltage Supply Class:
S2

PoE Class:
0, 1, 2, 3 and 4 (max 30W per port)

PoE Budget (option):
60 W derating of 1 W/ °C > 55 °C

Total Power Consumption
22 W (noPoE version)
100 W (PoE version)

Degree of protection
Standard IP40
Optional IP65

Analog Input Module

The EKE-Trainnet® Analog Input Module (AIM) provides a standard interface for train-borne signals, featuring six configurable measurement channels for voltage and current inputs. For each channel, either a voltage or current input can be selected via software configuration. In addition, the module includes four measurement channels for frequency inputs.

Compliance

- EN 50126
- EN 50716
- EN 50155
- EN 45545

Key Features

Current & Voltage Measurement Channels

AIM has:

- 6 current channels
- 6 voltage channels, and
- 4 frequency input channels in two groups: two for the low-voltage tachometer measurements and two for the high voltage pulse width modulation measurements.

The current measurement channels can measure currents between -20 and +20 mA. The module has the capability to supply voltage to external sensors or to supply loop voltage to the current loops. The module's voltage measurement channels can measure voltages between -10 and +10 V.

Signal Measurements

The frequency input channels can measure frequencies up to 16 kHz, period, pulse width, pulse count and rotary decoding with direction indication (compatible with most industry standard tachometers).

32-Bit Embedded Processor

The module has its own 32-bit embedded processor that uses factory calibration values to convert the measurement value to mV or µA with great accuracy. The embedded processor also implements diagnostics functions and reports to the system CPU module if it suspects that the input values cannot be trusted.

Over Range Indications for Diagnostics

The resolution of the input A/D conversion is sign+12 bits, and the cut-off frequency of the hardware low pass filter is 10 Hz. Both current and voltage channels are capable of giving over range indications which can be used for diagnostic purposes.



EKE-Trainnet® Analog Input Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight: 160 g

Input Power:
5 V DC ± 5 % (700 mA typ. 3 A max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
500 000 h (AIM2505)
1 600 000 h (AIM5250)

I/O Connector: DIN41612-F48 (at front)

Host Interface: RS-485

Analog Inputs:
6 voltage channels, -10...10 V (Vin)
6 current channels, -20...20 mA (Iin)
Connected in 6 measurement channels.

Frequency Inputs:
4 input channels in 2 groups: 2 low-voltage tachometer, 2 high-voltage PWM

Isolation Group 1:
Voltage1 or Current1,
Voltage2 or Current2

Isolation Group 2:
Voltage3 or Current3
Voltage4 or Current4

Isolation Group 3:
Voltage5 or Current5
Voltage6 or Current6

Isolation Group 4:
Frequency1, Frequency2, Frequency3, Frequency4

Frequency Input Voltage Range:
0...36 V DC

Frequency Input Threshold level:
Each channel has its own software configurable threshold level, fixed hysteresis

Frequency Input Range:
DC to 16 kHz, 0.25 Hz resolution

Frequency Input Pulse Width Measurement:
0.1 % resolution

Tachometer Supply Outputs (Vout):
15 V DC, max. 100 mA

High Speed Analog Input Module

The EKE-Trainnet® High Speed Analog Input module (HSA) provides a standard interface for analog input signals. Signals can be current (-20...+20mA) and voltage (-10...10V). Separate channels are provided for frequency signal measurements with the capability to measure pulse-width ratio (PWM).

Compliance

- ✓ SIL 2
- ✓ EN 50126
- ✓ EN 50716
- ✓ EN 50129
- ✓ EN 50155
- ✓ EN 45545

Key Features

Four Isolation Groups

The HSA module has:

- ✓ 4 current channels and
- ✓ 4 voltage channels

divided into four isolated groups. 4 frequency channels are also available.

Signal Measurements

The four frequency input channels can measure frequencies up to 16 kHz, period, pulse width, pulse count and rotary decoding with direction indication (compatible with most industry standard tachometers).

Current & Voltage Measurement Channels

The four current measurement channels can measure currents between -20 and +20 mA. The module has the capability to supply voltage to external sensors or to supply loop voltage to the current loops The module 4 voltage measurement channels can measure voltages between -10 and +10 V.

Over Range Indications for Diagnostics

The resolution of the input A/D conversion is sign+12 bits, and the cut-off frequency of the hardware low pass filter is 150 Hz. Both current and voltage channels are capable of giving over range indications which can be used for diagnostic purposes.

32-Bit Floating Point Digital Signal Processor

The module has its own 32-bit Floating point digital Signal Processor (DSP) for a high signal processing capability and uses factory calibration values to convert the measurement value to mV or µA with great accuracy. The embedded DSP processor can be used to perform signal processing tasks like filtering. It also implements diagnostics functions and reports to the system CPU module if it suspects that the input values cannot be trusted.

Options

Non-SIL Version

The HSA module is certified to SIL 2. A non-SIL version is also available.



EKE-Trainnet® High Speed Analog Input Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
170 g

Input Power:
5 V DC ± 5 % (1 A typ. 3 A max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
420 000 h

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
VME Bus A24/D08/D16 Slave

Analog Input:
4 voltage channels, -10...10 V (Vin)
4 current channels, -20...20 mA (Iin)

Frequency Input:
4 input channels

Frequency Input Voltage Range:
0...36 V DC (HSA2620A)

Frequency Input Current Range:
0...24 mA (HSA2620B)

Analog Input Bandwidth:
150 Hz (HSA2620A, HSA2620B)
3 kHz (HSA5530A)

Frequency Input Threshold level:
Each channel has its own software configurable threshold level, fixed hysteresis

Frequency Input Range:
DC to 16 kHz, 0.25 Hz resolution

Frequency Input Pulse Width Measurement:
0. 1% resolution

Group-wise Output Supply (e.g. for Tachometer):
15 V DC, max. 70 mA each

Analog Output Module

The EKE-Trainnet® Analog Output Module (AOM) provides a standard interface for analog output signals. The EKE-Trainnet® AOM receives instructions from the train computer's Central Processing Unit (CPU) module and sends analog signals to sub-systems accordingly.

Compliance

- EN 50126
- EN 50716
- EN 50155
- EN 45545

Key Features

Channel Configuration

By default, two channels are configured as bipolar voltage output and two digital pulse width modulation outputs (voltage mode). The channel configurations can easily be changed from current mode to voltage mode and vice-versa, depending on your needs.

32-Bit Embedded Processor

The module has its own 32-bit embedded processor to manage serial communication with the train computer's CPU module. The embedded processor also implements diagnostic functions and reports to the train computer's CPU module if it suspects that the output values cannot be trusted.

Calibration and Diagnostics

Each output channel has parameters for calibration (factory set), and diagnostics to detect short or open in current loop and short in voltage output. Outputs are factory calibrated.

Four Isolation Groups

Four individually isolated analog output groups are available. All groups are identical and have four analog output channels. Each group has two PWM outputs.

Internal Current Loop Supply Voltage

A current loop supply voltage is provided internally requiring no need for an external power supply. Configurable ranges are from 4 to 20 mA for current and from -10 to +10 V for voltage.



EKE-Trainnet® Analog Output Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
165 g

Input Power:
5 V DC ± 5 % (1 A typ., 3A max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
910 000 h (AOM2610)

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
RS-485

Current Output Mode:
4...20 mA
1 µA resolution

Voltage Output Mode:
-10...+10 V
1 mV resolution

Analog Output Channels:
16 (in 4 groups of 4 isolated channels)

PWM Outputs:
8 (2 in each of the 4 groups)
Selectable base frequency 10 Hz to 10 kHz Pulse width adjustable with 0.1 % resolution
Output level 15 V

Digital Input/Output Module

The EKE-Trainnet® Digital Input/Output module (DIO) provides a standard interface for managing digital input and output signals. Separate channels are provided for frequency measurements, which can be used for tachometer inputs.

Compliance

- SIL 2
- EN 50126
- EN 50716
- EN 50129
- EN 50155
- EN 45545

Key Features

Dedicated Channels

The DIO modules features:

- 24 dedicated input channels and
- 8 channels that can be configured as either inputs or outputs.

The DIO2908 and DIO2378 models allow up to four outputs to be connected in parallel for increased current capacity. Additionally, the DIO2378 includes 2 frequency input channels.

Emergency Output Short Circuit Protection

The module includes emergency output short circuit protection. If the output current exceeds its maximum value, the short circuit protection shuts down the output. The status of this protection is communicated to the train computer's CPU. In the event of communication failure, the output channel defaults to an emergency state defined in the application software, with individual settings for each channel.

Diagnostic Functions

Each input channel includes a common test circuit for real-time diagnostics, both at power-on and during normal operation. The train computer's CPU can access the input and output states, as well as configuration parameters such as wetting current and threshold levels.

32-Bit Embedded Processor

The module has its own 32-bit embedded processor to manage serial communication with the train computer's CPU module. The embedded processor also implements diagnostic functions and reports to the train computer's CPU module if it suspects that the output values cannot be trusted.

Two Separate Group Voltage

Two separate group voltages with common ground allow split behind two circuit breakers, or having a mixed voltage system. Input and Input/Output channels are split evenly between group voltage 1 and 2. Please note that the groups are not isolated from each other.

Options

Non-SIL Version

The DIO2378 module is non-SIL, while the DIO2908 module is SIL 2 certified.



EKE-Trainnet® Digital Input/Output Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
140 g

Input Power:
5 V DC ± 5 % (500 mA max., 200 mA typ.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
2 360 000 h (DIO2378)
1 450 000 h (DIO2908) (SIL 2)

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
RS-485

Input-only Channels: 24

Input/Output Channels: 8

Frequency Input Channels:
2 (DIO2378)

Frequency/High Speed Input Range:
0...32 kHz, resolution 1 Hz

Battery Voltage Monitor:
UIC-556 (DIO2378)

Temperature Measurement:
Yes, local CPU

Input Type: Current sink

Output Type:
High side (FET) from battery voltage

Voltage Drop:
Max. 2 V

Input/Output Voltage Range:
16.8...137.5 V DC (14...154 V DC, 1 sec)

Threshold levels:
Software selectable, both positive and negative. Either V or % Vin

Wetting Current:
0 ... 23 mA, Programmable (DIO2378)
20 mA (DIO2908)

Output Current:
1.0 A Continuous

Digital Relay Output Module

The EKE-Trainnet® Digital Relay Output Module (DRO) provides standard relay contacts. It receives commands from the train computer's central processing unit (CPU) module and controls the output relays accordingly. Relay contacts are accessible via the DIN41612 connector on the module's front.

Compliance

- EN 50126
- EN 50716
- EN 50155
- EN 45545

Key Features

Emergency States

The DRO modules are able to take up presetoutput states in case of emergency. If the communication between the train computer's CPU module and the DRO fails, the relays go to emergency states that you can define in the application software. When the DRO is unpowered, all relays are in "Normally Closed" state. Emergency states should be defined to match the wiring to "Normally Open" (NO) or "Normally Closed" (NC) contacts.



EKE-Trainnet® Digital Relay Output Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
190 g

Input Power:
5 V DC ± 5 % (500 mA max., 200 mA typ.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
1 000 000 h (DRO2342)

Relay electrical endurance:
500 000 operations (at 24 V DC 3.0 A resistive load)

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
RS-485

Number of Channels:
8 (DRO2342)

Relay Type:
Change over (SPDT) AgNi 0.15 Contacts (DRO2342)

Maximum Switching Current vs Voltage:
24 V DC 3.5 A (DRO2342)
48 V DC 1.4 A (DRO2342)
110 V DC 0.5 A (DRO2342)

Temperature Measurement:
Yes, local CPU

Pt100/Pt1000 Temperature Sensor Input Module

The EKE-Trainnet® PTI modules are temperature sensor input modules. They acquire the signals from the Pt100 (PTI2037) or Pt1000 (PTI3593) temperature sensors (resistance measurement) and convert them into temperature values expressed in degrees celsius.

Compliance

- SIL 2
- EN 50126
- EN 50716
- EN 50129
- EN 50155
- EN 45545

Key Features

Six Measurement Channels

The PTI has six measurement channels which can measure temperature within the -110 to +325 ° C range with over range indication. Each channel is isolated from each other, thus a fault in one channel will not affect the others. The temperature sensors can be connected using 2-wire, 3-wire or 4-wire schemes providing flexibility.

Diagnostic Functions

The embedded processor also implements diagnostic functions and reports to the train computer's CPU module if it suspects that the input values cannot be trusted. The PTI detects the temperature sensor as well as cable faults and input channel faults.

High Measurement Resolution

The measurement resolution is 0.01 °C, with accuracy better than ±1 °C within the measurement range.

32-Bit Embedded Processor

The module has its own 32-bit embedded processor that uses factory calibration values to convert measurement values to degrees Celsius with the best possible accuracy.



EKE-Trainnet® PTI Temperature Input Sensor Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
160 g

Input Power:
5 V DC ± 5 % (310 mA typ., 400 mA max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
2 490 000 h

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
RS-485

Input Channels:
6 Pt100 sensors (PTI2037)
6 Pt1000 sensors (PTI3593)

Connection Type:
2-wire, 3-wire or 4-wire

Measurement Range:
-110 °C to + 325 °C

Resolution:
0.01 °C

Accuracy:
Better than ± 1 °C (3- or 4-wire)

Temperature Sensor Input Module

The EKE-Trainnet® TSI is a temperature sensor input module. It acquires the signals from the Pt100, Pt1000 or NTC temperature sensors (resistance measurement) and converts them into temperature values expressed in degrees celsius. The used sensor type can be configured by software.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Three Sensor Types

Three sensor types are supported: NTC thermistor, Pt100 and Pt1000. Please note that the same sensor must be same for all six channels: Measurement range is:

- ✓ NTC thermistor:-25 to +125 °C
- ✓ Pt100 and Pt1000 sensors: -110 to +325 °C

All sensor types have an over range indication. The temperature sensors can be connected using 2-wire, 3-wire or 4-wire schemes providing flexibility.

Six Measurement Channels

Six measurement channels are available. Each channel is isolated from each other, thus a fault in one channel will not affect the others.

32-Bit Embedded Processor

The module has its own 32-bit embedded processor that uses factory calibration values to convert measurement values to degrees Celsius with the best possible accuracy.

Diagnostic Functions

The embedded processor also implements diagnostic functions and reports to the train computer's CPU module if it suspects that the input values cannot be trusted. The TSI detects the temperature sensor as well as cable faults and input channel faults.

High Measurement Resolution

The measurement resolution is 0.01 °C, with accuracy better than ±1 °C within the measurement range.



EKE-Trainnet® Temperature Input Sensor Module

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
160 g

Input Power:
5 V DC ± 5 % (310 mA typ., 400 mA max.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
2 490 000 h

I/O Connector:
DIN41612-F48 (at front)

Host Interface:
RS-485

Input Channels:
6 NTC, Pt100 or Pt1000

Connection Type:
2-wire, 3-wire or 4-wire

Measurement Range:
-25 °C to + 125 °C for NTC
-110 °C to + 325 °C for Pt100 and Pt1000

Resolution:
0.01 °C

Accuracy:
Better than ± 1 °C (3- or 4-wire)

Power Supply Unit For VME

The EKE-Trainnet® Power Supply Unit for VME (PSV) is a highly reliable switching mode power supply module. It is specifically designed for VME Bus systems and only works with the EKE-Trainnet® Power Input Unit (PIU) module which takes power from the train batteries and transfers it to the EKE-Trainnet® PSV module.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Prevent Component Damage

Built-in circuitry detects and controls the input current in the event of a power-up. Controlling the in-rush current prevents component damage.

Grace Supply Function

The grace supply function enables the module to continue working for a short period of time when the power is cut off. The PSV module withstands 10 ms input power breaks without disrupting any of the normal operations (Voltage Supply Interruption as per EN 50155 Class S2 and Supply change over as per EN 50155 Class C1). If the break lasts longer, the power supply is capable of supplying a +5 V DC output during a minimum of 100 ms after the input power is lost (80 ms for PSV1133).

Event Diagnostics

The ACFAIL and SYSRESET signals are activated to ensure proper shutdown of the systems. This feature allows the system to record subsystem and module states during the shutdown, thus enabling event diagnostics.

Current Consumption Minimized

When the input supply voltage is below the set minimum value, the current consumption is minimized in order to prevent the deep discharge of train batteries.



EKE-Trainnet® Power Supply Units for VME

Technical Specifications

Dimensions (W x H x D):
8 TE x 3 U x 160 mm

Weight:
500 g
(PSV1133, PSV2513, PSV1513, PSV1018)
470 g (PSV3034)

Temperature Range (operational)
-40 °C...+70 °C

MTBF (40 °C ambient temperature)
700 000 h (PSV1133, PSV3034)
690 000 h (PSV2513, PSV1513, PSV1018)

Input Voltage
24 V DC (PSV1133)
36 ... 52 V DC (PSV2513)
48 ... 110 V DC (PSV3034)
72 V DC (PSV1513)
110 V DC (PSV1018)

Input Voltage Range (V DC)
16.8 ... 30.0 (PSV1133)
25.2 ... 65.0 (PSV2513)
33.6 ... 143 (PSV3034)
50.4 ... 90.0 (PSV1513)
77 ... 138 (PSV1018)

Input Fluctuation (1s) (V DC)
14.4 ... 33.6 (PSV1133)
21.6 ... 72.8 (PSV2513)
31.2 ... 154 (PSV3034)
43.2 ... 100.8 (PSV1513)
66 ... 154 (PSV1018)

Input Current (<0.1 s/1 s)
5 A max (PSV1133)
4 A max (PSV2513)
3 A max (PSV3034, PSV1513)
2 A max (PSV1018)

Input Power
80 W max (PSV1133)
75 W max (PSV2513)
100 W max (PSV3034, PSV1513, PSV1018)

Supported Supply Interruption (S2)
10 ms

Efficiency
> 78%

Output Power
60 W (PSV1133)
54 W (PSV2513)
80 W (PSV3034, PSV1513. PSV1018)

Output Voltage 5 V DC

Maximum Output Current
8.5 A (PSV1133)
8.5 A (PSV2513) *
13 A (PSV3034, PSV1018)
12 A (PSV1513)

Minimum Output Current
0.5 A
(PSV1133, PSV2513, PSV1513, PSV1018)
0.1 A (PSV3034)

Ripple/noise p-p
50 mVpp (<20 MHz)

Output Voltage Hold-on at nom. load
100 ms

Output Voltage
5.15 V DC ± 2 % (at nominal line, nominal load)
5.00 V DC min. (at line range, load range)

Output Voltage 12 V DC

Maximum Output Current
1.25 A (PSV1133)
2.0 A (PSV2513)*
1.25 A (PSV3034, PSV1018, PSV1513)

Ripple/noise p-p
50 mVpp (<20 MHz)

Output Voltage
12.15 V DC ± 2 % (at nominal line, nominal load)
11.80 V DC min. (at line range, load range)

*****Total output loading must not exceed output power*

Power Input Unit

The EKE-Trainnet® Power Input Unit (PIU) is used to pass train battery voltage to the EKE-Trainnet® Power Supply Unit for VME (PSV) module, which powers EKE-Trainnet® systems across the train.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Range of Input Voltages Available

The EKE-Trainnet® PIU input voltages can be:

- ✓ 24 V DC (PIU2056)
- ✓ 24 to 36 V DC (PIU4233)
- ✓ 36 to 52 V DC (PIU2510)
- ✓ 48 to 110 V DC (PIU4234)
- ✓ 72 V DC (PIU2280)
- ✓ 110 V DC (PIU2219)

Please note that an EKE-Trainnet® PSV with a similar input voltage must be selected.

Options

Isolated DC Converter Output

An isolated DC/ DC converter may be installed as an option to provide power to any external equipment. If required, the output voltage may be switched off during the system reset in order to force hardresets of external equipment. Currently available PIU modules with DC Converter Output are listed under Technical Specifications. We can also provide you with other versions upon request.

Vehicle Identification Unit

The EKE-Trainnet® Vehicle Identification Unit (VIU) has the functionalities of the PIU, and in addition, provides an interface for Electronic Serial Number (ESN). An ESN dongle can be connected to the VIU for device identification and configuration. There are VIU versions for 24 V DC (VIU3688) and 110 V DC (VIU2336) input voltages.



EKE-Trainnet® Power Input Unit

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
100 – 150 g

Input Voltage:
24, 24–36, 36–52, 48–110, 72 or 110 V DC available
Others on request

Temperature Range (operational):
–40 °C...+70 °C

MTBF (40 °C ambient temperature):
38 570 000 h

Output Voltage (optional):
12 V DC (from PSV)

Technical Data DC Converter option

Weight:
350 g

Input Voltage:
24 or 110 V DC available Others on request

MTBF (40 °C ambient temperature):
2 400 000 h

Output Voltage and Power:
See below

Vehicle Identification Unit (VIU)

Weight:
155 g

Input Voltage:
110 V DC

MTBF (40 °C ambient temperature):
5 290 000 h

Available PIU Modules with DC Converter Output

Input Voltage
24 V DC (PIU2034)
110 V DC (PIU2023)
110 V DC (PIU3394)

Output Voltage
16 V DC (PIU2034. PIU2023)
24 V DC (PIU3394)

Output Power
17 W (PIU2034, PIU2023)
25 W (PIU3394)

Power Supply Unit

The EKE-Trainnet® Power Supply Unit (PSR) is a highly reliable switching mode power supply module. It takes power directly from the train batteries. Primarily designed to power the EKE-Trainnet® Remote Input/Output System, it can also be used for some simple VME Bus systems.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Range of Input Voltages Available

The PSR can be operated with 48 to 110 V DC (PSR3032) coach supply voltage and provide 5 V DC output power. The input supply is provided directly from the train battery to the PSR module through a 2W2C D-Sub connector.

Power Interruption Protection

Up to 10 ms of input power breaks can be withstood without disrupting any of the normal operations (Voltage Supply Interruption as per EN 50155 Class S2 and Supply change over as per EN 50155 Class C1). In case of a longer break in the power supply, the SYSRESET signal is activated to ensure proper shutdown of the systems.

Prevent Component Damage

Built-in circuitry detects and controls the input current in the event of a power-up. Controlling the in-rush current prevents component damage.

Current Consumption Minimized

When the input supply voltage is below the set minimum value, the current consumption is minimized in order to prevent the deep discharge of train batteries.

Technical Specifications

Dimensions (W x H x D):
8 TE x 3 U x 160 mm

Weight:
410 g

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
800 000 h

Input Voltage:
10 V DC

Input Voltage Range (V DC):
77 ... 138

Input Fluctuation (1s) (V DC):
66 ... 154

Input Current (<0.1s/1s):
4 A max.

Supported Supply Interruption (S2):
10 ms

Efficiency:
> 80% (typically 85%)

Output Power:
80 W

Output Current: 16 A max.

Minimum Output Current:
0.1 A min.

Ripple/noise p-p:
50 mVpp (<20 MHz)

Output Voltage
At nominal line, nominal load:
5.15 V DC ± 2 %

At line range, load range:
5.05 V DC min



EKE-Trainnet® Power Supply Unit

Vehicle Identification Unit

The EKE-Trainnet® Vehicle Identification Unit (VIU) is a module designed for vehicle identification and configuration management. It provides an interface for Electronic Serial Number (ESN) functionality, enabling train-specific configuration data to be stored and retrieved.

Compliance

- EN 50155
- EN 45545
- EN 6000
- EN 60068
- IEC 61373

Key Features

Electronic Serial Number (ESN) Interface

The VIU provides an interface for the Electronic Serial Number (ESN) module. The ESN stores vehicle-specific identification and configuration data that can be read by the on-board system during operation.

Vehicle Identification

Vehicle identification is enabled within the Train Control and Management System. Vehicle identification data can be used by on-board applications to determine the correct vehicle configuration and operating parameters.

Configuration Data Storage

Configuration data is stored in the ESN module connected to the VIU. This allows vehicle-specific settings to be retained independently of other system modules and transferred when equipment is replaced.

Support for Fleet Configuration Management

The VIU supports the management of train and vehicle configuration data by providing a dedicated hardware interface for identification and configuration information.

Technical Specifications

Dimensions (W x H x D):
4 TE x 3 U x 160 mm

Weight:
155 g

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
5 300 000 h

Input Voltage:*
110 V DC (VIU2336)
24 V DC (VIU3688)

Maximum Continuous Input Current:*
1 A (VIU2336)
6 A (VIU3688)

P1 Download and Electronic Serial Number (ESN) Interface HD-15

D-sub 15-pin, female

P3 Connector Supply Input:
Female D 2W2C

** The VIU module does not itself draw current from this supply. Its internal circuitry is powered from the PSV module.*



EKE-Trainnet® Vehicle Identification Unit

Electronic Serial Number

The EKE-Trainnet® Electronic Serial Number (ESN) module is a dedicated memory module used for vehicle identification and configuration management. It stores train-specific configuration data that is read by the on-board system during start-up. Primarily designed for use with the EKE-Trainnet® Vehicle Identification Unit (VIU), it supports efficient commissioning, maintenance and equipment replacement activities.

Key Features

Vehicle Configuration Storage

The ESN stores vehicle-specific configuration data that can be accessed by the on-board system. This allows configuration information to remain associated with the vehicle rather than individual system modules.

Electronic Identification

The ESN provides electronic identification for the vehicle or train application. Identification data can be used by on-board systems to determine the correct configuration during operation.

Configuration Data Transfer

Configuration data is programmed into the ESN and can be transferred with the module when equipment is replaced. This simplifies the retention of vehicle-specific settings.

System Start-Up Support

During system start-up, configuration data stored in the ESN is read by the on-board system. This enables the correct vehicle configuration to be established automatically.

Protected Configuration Data

The ESN stores configuration as a protected binary data package, helping to preserve the integrity of vehicle configuration information.

ESN Programming Tool

Stores train-specific identification and configuration data, enabling engineers to read, edit and write settings that remain associated with the vehicle rather than individual hardware modules.

Options

Choose the Right Memory Capacity

ESN2218 is equipped with a 4 kbit EEPROM memory, whereas ESN4463 is equipped with a 20 kbit EEPROM memory.



EKE-Trainnet® Electronic Serial Number Module

Compliance

- EN 50155
- EN 45545
- EN 60068
- EN 6000
- IEC 61373

Technical Specifications

Dimensions (W x H x D):
15 mm x 37 mm x 58 mm

Weight:
60 g

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
66 410 000 h (ESN2218)
84 540 000 h (ESN4463)

HD D-sub 15 Pin M:
1

Rugged Memory Module

The EKE-Trainnet® Rugged Memory Module (RMM) is a highly protected memory module to be used as part of the EKE-Trainnet® Event Recorder. As an essential part of the Train Event Recorder, the RMM collects and stores vital train-borne information. Depending on your needs, it can record data such as audio and video.

Compliance

- EN 50155
- EN 45545
- IEEE 1482.1-1999
- IEC 62625-1:2013 GM/ RT2472:2014 EEIG 97E461-3.1998

Key Features

Store Analog and Digital Signal Data

Both analog and digital signal data are stored for a set period. Data can be accessed anytime using the EKE-Trainnet® TIP Tool for performance analysis and optimisation.

Easy Access to Data

Data from the EKE-Trainnet® RMM can be retrieved through the Ethernet port on the module's front panel, as well as via the VME back plane, from added interface modules, and the RS-485 port.

High Level of Data Protection

The RMM is designed for high data protection. Its crash-proof construction ensures protection against fire, magnetic fields, liquids, and mechanical stress from impacts or continuous pressure. The memory board is housed in a fire-resistant insulation block and enclosed in a hermetically sealed steel case.

Record Video Streams

Video streams from on-board digital video recorders (DVR) or directly from cameras can be recorded. For easy data retrieval, the stream format can be made compatible with your CCTV software or any other video analysis software that you use. Alternatively, we can develop customized video analysis tools to fit your needs.

Standalone or Integrated Event Recording

When paired with an EKE-Trainnet® 3U 84TE Rack and power supply (typically EKE-Trainnet® PSV and PIU modules), a fully functional standalone event recorder is created. The RMM can also be integrated into a larger EKE-Trainnet® system, complementing Gateways, Vehicle Control Units, or Train Control and Management Systems.



EKE-Trainnet® Rugged Memory Module

Technical Specifications

Dimensions (W x H x D):
32 TE x 3 U x 160 mm

Weight:
7 Kg

Input Power:
5 V DC ± 5 % (1 A max., 0.5 A typ.)

Temperature Range (operational):
-40 °C...+70 °C

MTBF (40 °C ambient temperature):
890 000 h (2 GB)
870 000 h (16 GB)

Interfaces:
3 or 4 Isolated RS-485
1 x 10/100 Mbit/s Ethernet, M12 connector

Memory:
2 GB or 16 GB

Protection Levels

Ingress Protection (IP) rating:
68

Shock:
Up to 100 g / 10 ms and 55 g / 100 ms

Crush: Up to 110 kN for 5 minutes in all axes

Penetration:
Up to 23 kg drop from a height of 1.5 m focussed on an area of 30 mm^2

Temperature:
Up to 700 °C for 5 minutes and 650 °C for 30 minutes followed by 300 °C for 1 hour followed by 100 °C for 5 hours

Chemical Immersion:
Winter Diesel*, Summer Diesel*, Regular Water*, Salt Water*, Lubricating Oil*, Transformer Oil**, Hydraulic Oil**, Anti Fire Fighting Foam *, Refrigerant R134A**, Alcohol**, Anti Freeze**, Battery Acid **

** minimum for 48 hours*

*** minimum for 60 min*

Magnetic Field:
Current flow of 64 kA at rising rate of 107 A/s

Hydrostatic Pressure:
Immersion in salt water at a depth of 15 meters for 48 hours

Standard Racks

EKE-Trainnet® Standard Racks enable communication between modules via an integrated IEC 821 VME back plane bus. All modules take power from the power supply via the back plane. EKE-Trainnet® Racks are required with every EKE-Trainnet® System. EKE-Trainnet® modules only work when integrated into an EKE-Trainnet® Rack.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Different Sizes to Fit All

Different sizes are available to fit all needs. While 44TE and 84TE wide racks are commonly used, a 20TE wide rack is also available for smaller systems. When working with a high number of modules the 6U 84TE can be used. Other racks sizes are also available upon request.

Different Back Plane Options

The table below outlines the different back plane combinations. The number of free slots for both VME modules (CPUs, bus interface modules and HSA module) and I/O modules (except HSA) are mentioned. The space required for the power supply is already taken into account.

Technical Specifications

External Dimensions (W x H x D):

3U 20TE Rack:
158 mm x 133 mm x 215 mm (installation width 117 mm)

3U 44TE Rack:
280 mm x 133 mm x 215 mm (installation width 240 mm)

3U 84TE Rack:
483 mm x 133 mm x 215 mm (installation width 443 mm)

6U 84TE Rack:
483 mm x 266 mm x 215 mm (installation width 443 mm)

Weight:
3U 20TE Rack: 1.3 Kg
3U 44TE Rack: 2.3 Kg
3U 84TE Rack: 4 Kg
6U 84TE Rack: 7.3 Kg

Ingress Protection (IP) rating:
20

Options

Side Mounting for 20TE Racks

The 3U 20TE Racks can be sided mounted as an option, instead of the usual front mounting. Dimensions are then W127 x H133 x D306 mm.

Available VME and I/O Slots

Rack	VME only slots	VME or 1/O slots	I/O only slots	MTBF	Comments
3U 20TE	2	1	0	3 480 000 h	-
3U 44TE	1	7	0	1 830 000 h	-
3U 44TE RIOM	1	0	8	2 330 000 h	-
3U 84TE	4	6	8	980 000 h	-
3U 84TE Dual: Left	1	6	0	970 000 h	-
3U 84TE Dual: Right	2	6	0	970 000 h	The right back plane has an extra VME slot that usually remains unused
3U 84TE RMM	4*	7	0	1 440 000 h	28TE* reserved for RMM
6U 84TE	Select a 3U 84TE rack version		18	850 000 h	The other 3U fits 2 x 1U ESU



EKE-Trainnet® 3U 20TE Rack



EKE-Trainnet® 3U 44TE Rack



EKE-Trainnet® 3U 84TE Dual Rack



EKE-Trainnet® 6U 84TE RMM Rack

Flat Racks

The EKE-Trainnet® Flat Racks are designed to be used in trains where space is limited, for instance trams and light rail vehicles. They also provide an alternative to standard racks dimensions and mounting mechanisms. All modules take power from the power supply via the back plane. EKE-Trainnet® Racks are required with every EKE-Trainnet® System. EKE-Trainnet® modules only work when integrated into an EKE-Trainnet® Rack. Please note that EKE-Trainnet® Flat Racks are only available by request.

Compliance

- ✓ EN 50155
- ✓ EN 45545

Key Features

Variety of Small Sizes Available

Flat Racks come in 3U and 6U format, either as a 12TE or 16TE wide rack. Thanks to their small sizes, Flat Racks can be used in the vicinity of train systems, thus allowing savings by reducing cabling.

Different Back Plane Options

The number of free slots for both VME modules (CPUs, bus interface modules and HSA module) and I/O modules (except HSA) are mentioned in the table below. The space required for the power supply is already taken into account. Please note that in the 6U versions, there is no VME bus communication between the two rows of modules.

Technical specifications

External Dimensions (W x H x D):

3U 12TE Flat Rack:
87 mm x 229 mm x 215 mm

3U 16TE Flat Rack:
107 mm x 229 mm x 215 mm

6U 12TE Flat Rack:
87 mm x 363 mm x 215 mm

6U 16TE Flat Rack:
107 mm x 363 mm x 215 mm

Weight:
3U 12TE Rack: 1.21 Kg
3U 16TE Rack: 1.32 Kg
6U 12TE Rack: 2.15 Kg
6U 16TE Rack: 2.33 Kg

Ingress Protection (IP) rating:
20

Available VME and I/O Slots

Rack	VME only slots	VME or I/O slots	I/O only slots	MTBF
3U 12TE	0	1	0	8 150 000 h
3U 16TE	0	2	0	5 680 000 h
6U 12TE	0	4	0	3 140 000 h
6U 16TE	0	5	1	2 440 000 h





EKE-Trainnet® 3U 12TE Flat Rack



EKE-Trainnet® 3U 16TE Flat Rack



EKE-Trainnet® 6U 12TE Flat Rack



EKE-Trainnet® 6U 16TE Flat Rack

Notes



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